Antihemolytic activity of *Stipagrostis pungens* (Desf.) De Winter extracts: proposing a novel molecular mechanism of erythrocyte membrane stabilization

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Introduction

Oxidative stress is widely recognized as a major contributor to lipid peroxidation, protein oxidation, and erythrocyte membrane damage, ultimately leading to premature hemolysis and various hematological disorders [1, 2]. Human red blood cells (RBCs) are continuously subjected to endogenously generated reactive oxygen species (ROS) and mechanical shear stress throughout their 120-day lifespan [3, 4]. Although plant polyphenols are widely recognized for their free radical scavenging abilities [5], the precise molecular mechanisms underlying their direct protective effects on erythrocyte membrane structural proteins remain largely unexplored [6]. *Stipagrostis pungens* (Desf.) De Winter (Poaceae) is traditionally used in southern Algeria as a remedy for anemia [7].

Objectives

Evaluate the anti-hemolytic and membrane-protective capacity of *S. pungens* aqueous and hydro-acetone extracts under AAPH-induced oxidative stress and hypotonic conditions [6,7].
Characterize key phenolic compounds using LC-ESI-MS/MS [8].
Propose a novel molecular mechanism of erythrocyte membrane stabilization through *in silico* docking against structural proteins (Band 3 and Ankyrin) [6, 9].

Materiel & Methods

Plant Material & Extraction

Aerial parts of *S. pungens* were collected from Laghouat (Algerian Saharan steppe) [7]. Dried plant material was ground into a fine powder and extracted by maceration (24 h at room temperature) using distilled water and hydro-acetone (70:30, v/v) [7].

LC-ESI-MS/MS Phytochemical Profiling

Chromatographic separation was performed on a C18 column using a Shimadzu LC-MS-8050 system operated in ESI-negative Multiple Reaction Monitoring (MRM) mode with authentic reference standards [8].

Anti-Hemolytic & Membrane Stability Assays

Human erythrocytes isolated from healthy volunteers were pre-incubated with plant extracts (0.5–8 mg/mL) and exposed to 75 mM AAPH (peroxyl radical generator) at 37 °C [6, 7].
Turbidity / Membrane Integrity: Measured at 620 nm following cell fixation [6].
Hemoglobin Release & Methemoglobin: Quantified at 412 nm and 540 nm, respectively [6, 7].

In Silico Molecular Docking

Molecular docking was executed using MOE 2019.01 software (Amber10 / MMFF94 force fields) against crystallographic targets retrieved from the Protein Data Bank: Cytoplasmic Band 3 (PDB ID: 1HYN), Transmembrane Band 3 (PDB ID: 4YZF), and Ankyrin (PDB ID: 3KBT) [10]. Protocol validation was confirmed by re-docking the 4KU native ligand (RMSD = 0.9291 Å, S = -8.2874 kcal/mol).

Results & Discussion

LC-ESI-MS/MS analysis annotated major phenolic constituents across the extracts.

Rosmarinic acid was identified as the predominant phenolic compound (16.605 mg/g DE in aqueous and 7.090 mg/g DE in hydro-acetone extract) The hydro-acetone extract specifically co-extracted higher concentrations of semi-polar flavonoid aglycones, including luteolin-7-*O*-glucoside (0.206 mg/g), epicatechin (0.049 mg/g), kaempferol (0.023 mg/g), diosmetin (0.015 mg/g), and biochanin A (0.012 mg/g).

Erythrocyte Protection Against Oxidative Hemolysis

Inhibition of Hemolysis (412 nm) & Methemoglobin (540 nm): Hydro-acetone extract demonstrated superior efficacy compared to aqueous extract, reducing absorbance at 412 nm down to 0.16 ± 0.01 at 4 mg/mL and preventing methemoglobin accumulation.

Membrane Preservation (620 nm): Turbidity values at 620 nm remained high in treated samples, reflecting preserved cellular resistance.

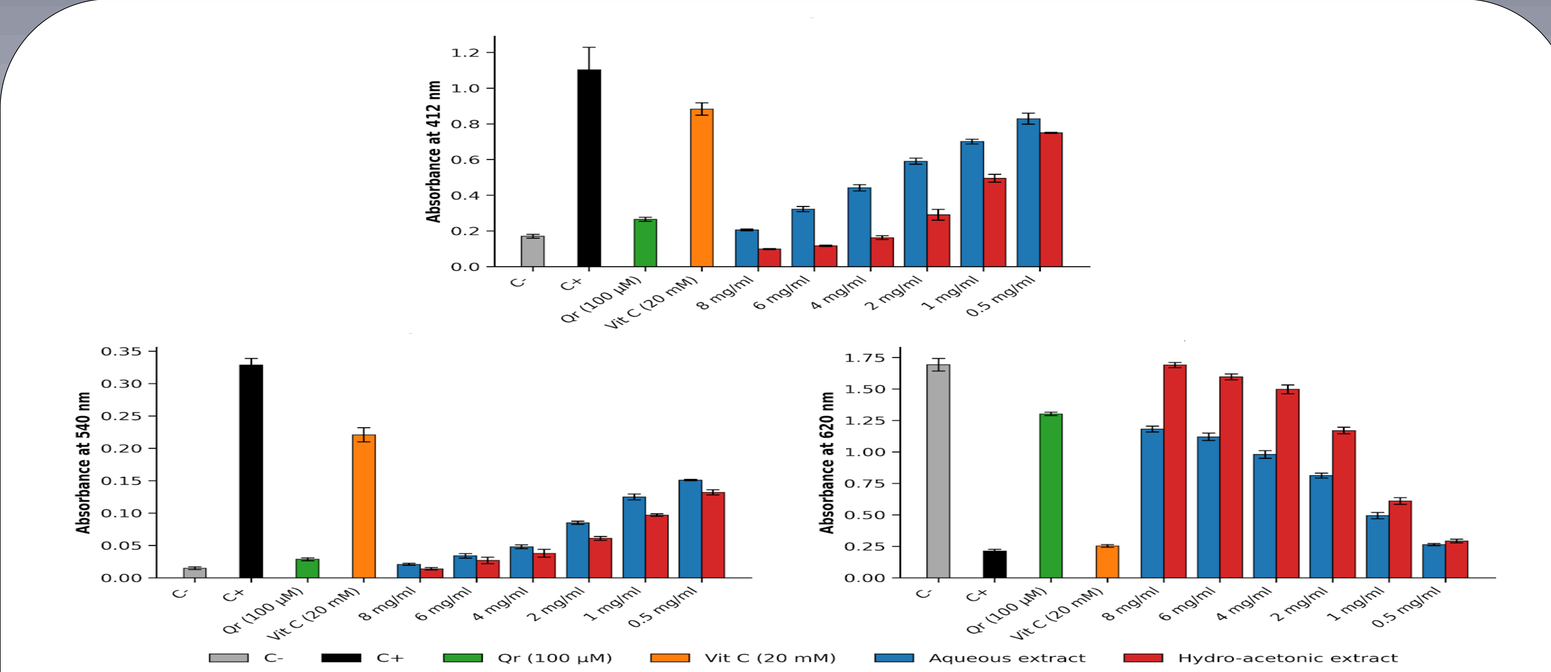


Fig.1. Concentration-dependent anti-hemolytic activity of *S. pungens* against AAPH-induced oxidative stress in human erythrocytes. Protection was assessed against positive (AAPH) and negative (PBS) controls. Quercetin (Q) and Vitamin C were used as reference standards.

Docking simulations revealed strong predicted affinities (S-scores: -6.11 to -8.48 kcal/mol, RMSD < 2.00 Å) between identified phenolics and the Band 3–Ankyrin cytoskeletal network.Cyto-Band 3 (1HYN): Luteolin-7-*O*-glucoside displayed an outstanding binding score at Site 2 (S = -8.4835 kcal/mol, RMSD = 1.7689 Å), forming H-donor bonds with ASP316 (2.94 Å, E = -1.7 kcal/mol) and SER194 (3.19 Å). Rosmarinic acid anchored to Site 2 (S = -7.4435 kcal/mol) via H-bonds with SER194 (3.14 Å) and ionic bonds with ARG233 (2.95 Å and 3.45 Å). Ankyrin (3KBT): Rosmarinic acid bound strongly to regulatory Site 1 (S = -7.0072 kcal/mol), forming H-bonds with TRP1705 (3.06 Å) and SER1715 (2.75 Å), plus ionic bonds with LYS1709 (3.00 Å, E = -4.5 kcal/mol) and LYS1731 (3.54 Å). Transmembrane Band 3 (4YZF): Lipophilic aglycones enriched in hydro-acetone extract targeted lipid-facing regions: diosmetin formed a π–cation interaction with LYS592 (4.46 Å; S = -6.9842, RMSD = 0.8841 Å) and π–π stacking with Ankyrin's PHE511 (2.58 Å; S = -6.1130, RMSD = 0.9291 Å), while biochanin A established an ionic bond with Band 3 s' ARG730 (3.25 Å, E = -3.0 kcal/mol; S = -6.4009, RMSD = 1.5149 Å) Physiological Safety: Importantly, ligands did not interact with the N-terminal acidic sequence (residues 1–30) of Cyto-Band 3 [9]. As this N-terminal region is essential for binding deoxyhemoglobin and glycolytic enzymes (aldolase, GAPDH, phosphofructokinase) [9], polyphenols stabilize structural integrity without impairing cellular respiration or energy metabolism.

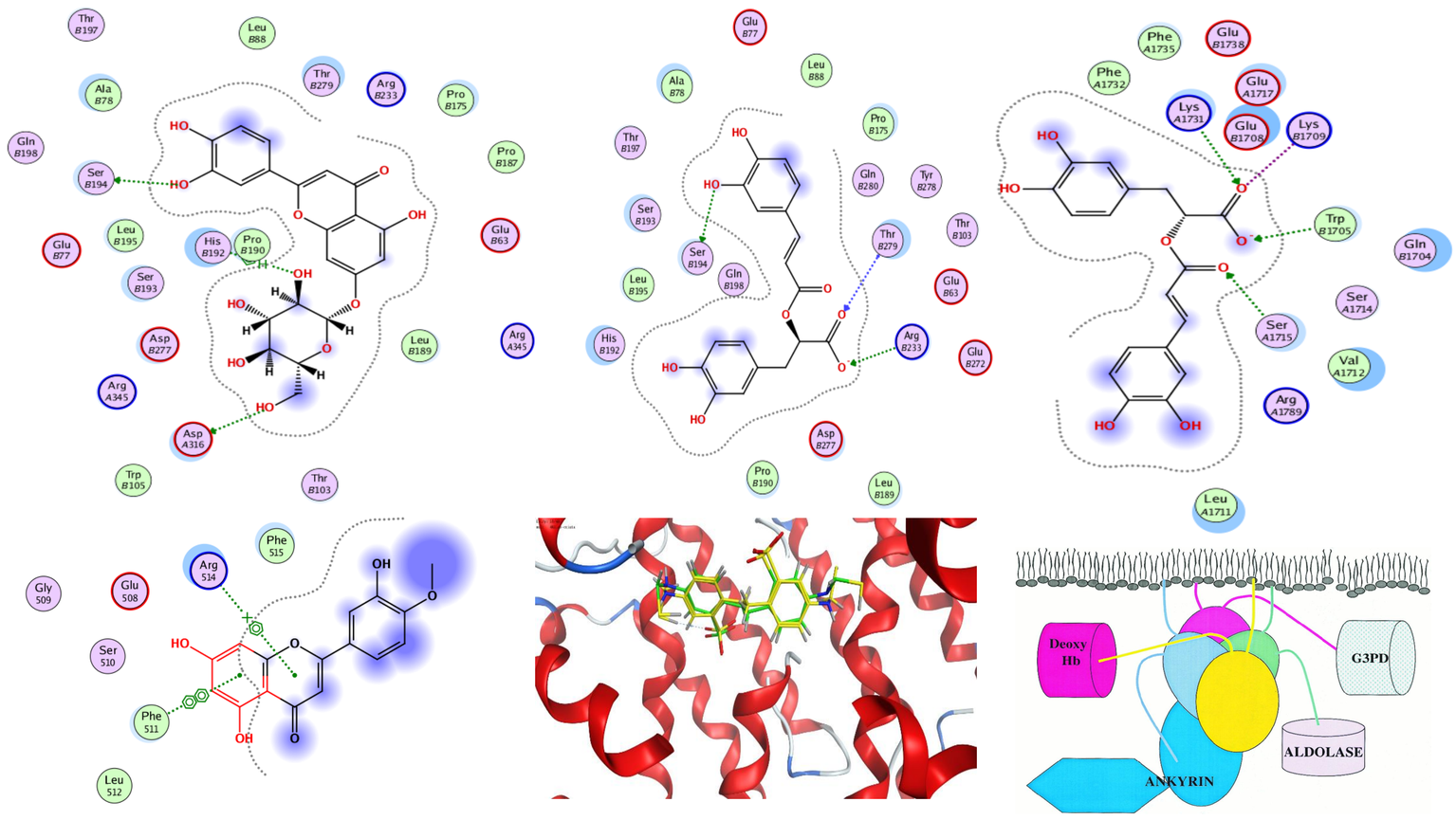


Fig.2. *In silico* docking targeting the Band 3–ankyrin complex. Predicted binding poses and allosteric stabilization of the erythrocyte Band 3 cytoplasmic domain.

Conclusion & Perspectives

This study provides that *Stipagrostis pungens* extracts exert anti-hemolytic activity through a possible dual-action mechanism, potential structural stabilization of Band 3 and Ankyrin proteins, preventing cytoskeletal uncoupling without interfering with physiological domains. These findings validate the traditional use of *S. pungens* and highlight its potential as a source of natural membrane-stabilizing agents for managing oxidative hemolytic disorders.

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